

EDUCATION	Massachusetts Institute of Technology, Cambridge, MA GPA: 5.0/5.0 B.S. in Mathematics, M.S. in Artificial Intelligence, 2021-2025 <i>Relevant coursework:</i> <table><tr><td>Deep Learning (G)</td><td>Complexity Theory (G)</td><td>Inference and Information (G)</td></tr><tr><td>GPU Programming (G)</td><td>Theory of Probability (G)</td><td>Embodied Intelligence (G)</td></tr><tr><td>Performance Engineering</td><td>Quantum Mechanics III</td><td>Field and Galois Theory</td></tr></table>	Deep Learning (G)	Complexity Theory (G)	Inference and Information (G)	GPU Programming (G)	Theory of Probability (G)	Embodied Intelligence (G)	Performance Engineering	Quantum Mechanics III	Field and Galois Theory
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Performance Engineering	Quantum Mechanics III	Field and Galois Theory								
CURRENT RESEARCH	Faster Optimizers via Modular Duality, advised under Jeremy Bernstein and Phillip Isola - Inspired Muon. TL;DR: for linear layers, update by $UV^\top$ for gradient $G = U\Sigma V^\top$.									
PUBLICATIONS	Old Optimizer, New Norm: An Anthology, NeurIPS 2024 (workshop) An Assessment of Model-on-Model Deception, ICLR 2024 (workshop) ANTN: Bridging Autoregressive Neural Networks and Tensor Networks for Quantum Many-Body Simulation, NeurIPS 2023 (poster) An Examination of Diversity in INFORMS Journal Editorial Boards, INFORMS Journal of Service Science 2021									
WORK EXPERIENCE	<table><tr><td> Khan Academy for Ukraine Co-Founder - Co-leading initiative to translate Khan Academy for every student in Ukraine. Center for Human-Compatible AI Researcher, Berkeley, CA - Implemented 40 new environments up to 4M states, researched long effective horizon RL. Commonwealth Fusion Systems Software Intern, Somerville, MA - Designed, implemented, and tested study of quench detection in the SPARC reactor. Sped up core physics algorithm by 23x using performance engineering. Amazon Web Services Software Intern, Arlington, VA - Designed, implemented, tested, and documented integrating AWS IAM into Apache Kafka for disconnected edge devices. Wrote 9000 lines of Java code, all going into production. Khan Academy Content Developer, Mountain View, CA - Created comprehensive set of 600 practice problems for Khan Academy's online multivariable calculus course, now serving over 10,000 unique users per month. - Interviewed subject matter experts, including Grant Sanderson from 3Blue1Brown. </td><td> Sep 2023 — Present Jun 2024 — Aug 2024 Jun 2023 — Aug 2023 Jun 2022 — Aug 2022 Jul 2019 — Sep 2020 </td></tr></table>	Khan Academy for Ukraine Co-Founder - Co-leading initiative to translate Khan Academy for every student in Ukraine. Center for Human-Compatible AI Researcher, Berkeley, CA - Implemented 40 new environments up to 4M states, researched long effective horizon RL. Commonwealth Fusion Systems Software Intern, Somerville, MA - Designed, implemented, and tested study of quench detection in the SPARC reactor. Sped up core physics algorithm by 23x using performance engineering. Amazon Web Services Software Intern, Arlington, VA - Designed, implemented, tested, and documented integrating AWS IAM into Apache Kafka for disconnected edge devices. Wrote 9000 lines of Java code, all going into production. Khan Academy Content Developer, Mountain View, CA - Created comprehensive set of 600 practice problems for Khan Academy's online multivariable calculus course, now serving over 10,000 unique users per month. - Interviewed subject matter experts, including Grant Sanderson from 3Blue1Brown.	Sep 2023 — Present Jun 2024 — Aug 2024 Jun 2023 — Aug 2023 Jun 2022 — Aug 2022 Jul 2019 — Sep 2020							
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AWARDS	TheCoderGames National Contest Winner — first place out of 300 nationwide (2018)									
LANGUAGES	<i>machine:</i> Python, C, JavaScript, Java, Verilog, Vim, Git, CUDA <i>human:</i> English, German, Spanish; beginning French and Hebrew									
INTERESTS	hiking, meditation, basketball, cheese, language learning, reading, and good math problems									